Auteur: Abdellah El Moussaoui
Studierichting: Informatica
Oefeningenreeks 5B nr. 3.1

$$\int_1^{+\infty} \frac{\ln x}{(x+1)^2} dx$$
$$= \lim_{a\to+\infty} \int_1^a \ln x\,(x+1)^{-2}\,dx$$

We weten dat: $\int_d^c udv = [uv]_d^c - \int_d^c vdu$

Laat:

$u = \ln x$, $dv = (x+1)^{-2}\,dx$

$du = \frac{1}{x}dx$, $v = -(x+1)^{-1}$

Dus:

$$\lim_{a\to+\infty} \int_1^a \ln x\,(x+1)^{-2}\,dx = \lim_{a\to+\infty} \left(-\left[\frac{\ln x}{x+1}\right]_1^a + \int_1^a \frac{1}{x\,(x+1)} dx \right)$$

We weten dat:

$$\frac{1}{x\,(x+1)} = \frac{A}{x} + \frac{B}{x+1}$$
$$\Rightarrow A = 1\ ,\ B = -1$$

$$\lim_{a\to+\infty} \int_1^a \ln x\,(x+1)^{-2}\,dx = \lim_{a\to+\infty} \left(-\left[\frac{\ln x}{x+1}\right]_1^a + \int_1^a \left(\frac{1}{x} - \frac{1}{(x+1)}\right) dx \right)$$
$$= \lim_{a\to+\infty} \left(-\left(\frac{\ln a}{a+1} - \frac{\ln 1}{2}\right) + [\ln x - \ln x + 1]_1^a \right)$$
$$= \lim_{a\to+\infty} -\frac{\ln a}{a+1} + \ln a - \ln(a+1) - \ln 1 + \ln(2)$$
$$= \lim_{a\to+\infty} -\frac{\ln a}{a+1} + \ln a - \ln(a+1) + \ln 2$$
$$= \lim_{a\to+\infty} -\frac{\ln a}{a+1} + \ln\left(\frac{a}{a+1}\right) + \ln 2$$
$$= \lim_{a\to+\infty} -\frac{\frac{1}{a}}{1} + \ln 1 + \ln 2$$
$$= \lim_{a\to+\infty} -\frac{1}{a}$$
$$= \ln 2$$

References